

Experiment-8

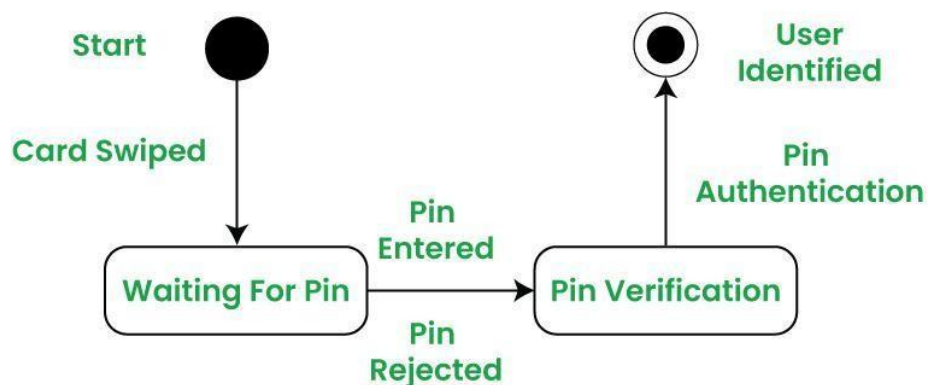
Objective- Statechart diagram

A Statechart diagram describes a state machine. State machine can be defined as a machine which defines different states of an object and these states are controlled by external or internal events.

Activity diagram explained in the next chapter, is a special kind of a Statechart diagram. As Statechart diagram defines the states, it is used to model the lifetime of an object.

- State Machine diagrams are also known as **State Diagrams** and **State-Chart Diagrams**. These both terms can be used interchangeably.
- A state machine diagram is used to model the dynamic behaviour of a class in response to time and changing external stimuli(events that cause the system to change its state from one to another).
- We can say that every class has a state but we don't model every class using State Machine diagrams.
- Let's understand the State Machine Diagram with the help **user verification** example:
- **Example:**

A State Machine Diagram for user verification



- The State Machine Diagram above shows the different states in which the verification subsystem or class exists for a particular system.
- **Basic Components and Notations of a State Machine Diagram**
- Below are the basic components and their notations of a State Machine Diagram:

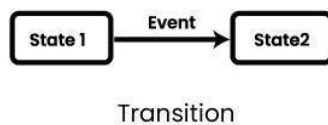
1. Initial state

We use a black filled circle represent the initial state of a System or a Class.



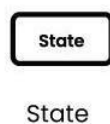
2. Transition

We use a solid arrow to represent the transition or change of control from one state to another. The arrow is labelled with the event which causes the change in state.



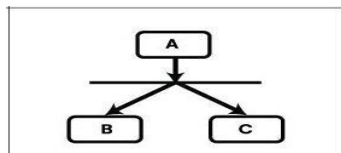
3. State

We use a rounded rectangle to represent a state. A state represents the conditions or circumstances of an object of a class at an instant of time.



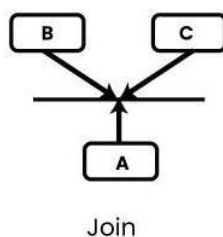
4. Fork

We use a rounded solid rectangular bar to represent a Fork notation with incoming arrow from the parent state and outgoing arrows towards the newly created states. We use the fork notation to represent a state splitting into two or more concurrent states.



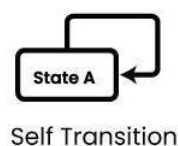
5. Join

We use a rounded solid rectangular bar to represent a Join notation with incoming arrows from the joining states and outgoing arrow towards the common goal state. We use the join notation when two or more states concurrently converge into one on the occurrence of an event or events.



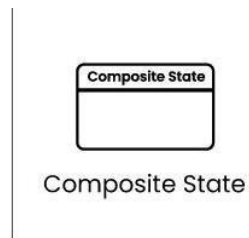
6. Self transition

We use a solid arrow pointing back to the state itself to represent a self transition. There might be scenarios when the state of the object does not change upon the occurrence of an event. We use self transitions to represent such cases.



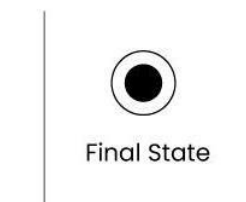
7. Composite state

We use a rounded rectangle to represent a composite state also. We represent a state with internal activities using a composite state.



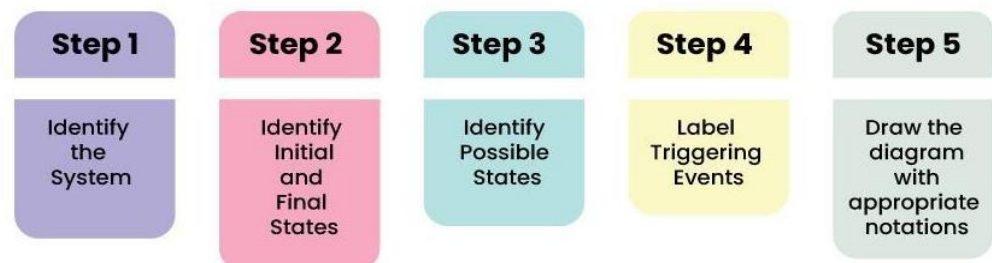
8. Final State

We use a filled circle within a circle notation to represent the final state in a state machine diagram.



How to draw a State Machine diagram in UML?

Below are the steps on how to draw the State Machine Diagram in UML:



Step 1: Identify the System:

- Understand what your diagram is representing.
- Whether it's a machine, a process, or any object, know what different situations or conditions it might go through.

Step 2: Identify Initial and Final States:

- Figure out where your system starts (initial state) and where it ends (final state).
- These are like the beginning and the end points of your system's journey.

Step 3: Identify Possible States:

- Think about all the different situations your system can be in.
- These are like the various phases or conditions it experiences.
- Use boundary values to guide you in defining these states.

Step 4: Label Triggering Events:

- Understand what causes your system to move from one state to another.
- These causes or conditions are the events.
- Label each transition with what makes it happen.

Step 5: Draw the Diagram with appropriate notations:

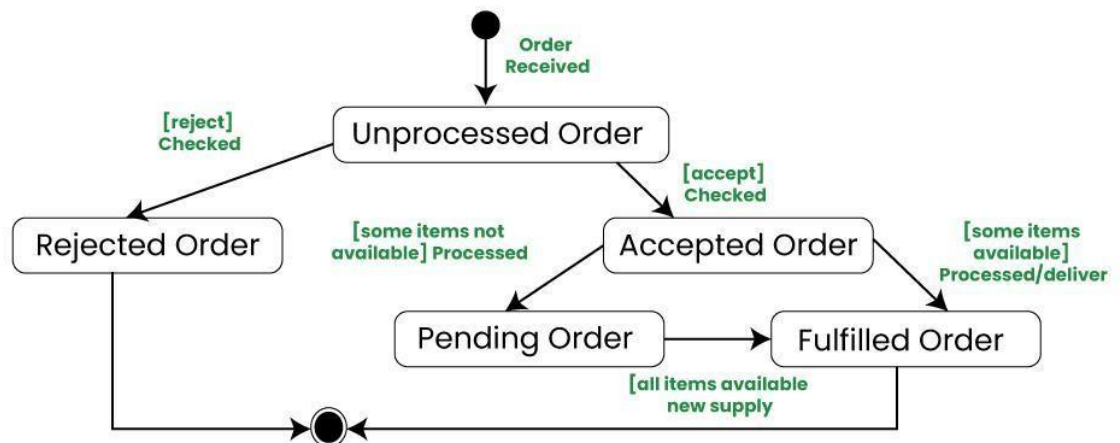
- Now, take all this information and draw it out.

- Use rectangles for states, arrows for transitions, and circles or rounded rectangles for initial and final states.
- Be sure to connect everything in a way that makes sense.

Example: Online Order System

- Let's understand State Machine diagram with the help of an example, ie for an **Online Order** :

State machine diagram for an online order



- The UML diagrams we draw depend on the system we aim to represent. Here is just an example of how an online ordering system might look like :
- On the event of an order being received, we transition from our initial state to Unprocessed order state.
- The unprocessed order is then checked.
- If the order is rejected, we transition to the Rejected Order state.
- If the order is accepted and we have the items available we transition to the fulfilled order state.
- However if the items are not available we transition to the Pending Order state.
- After the order is fulfilled, we transition to the final state. In this example, we merge the two states i.e. Fulfilled order and Rejected order into one final state.